# Vincent Ngo

Startup Founder and Author.
vincentngo.github.io | LinkedIn
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## Education

**Bachelor of Science in Computer Science**, *Virginia Tech* 2010 - 2014
Graduated May 2014 · Minor in Mathematics · GPA: 3.65/4.0 overall, 3.61/4.0 in major · Blacksburg, VA
Activities: Computer Science Community Service (CS2), Web Development Group, ACM, and hackathons including MHacks, PennApps, HackNC, and HackVT.

**Undergraduate Computer Science Coursework**, *Stanford University*
Completed undergraduate-level computer science courses while enrolled in high school.

## Positions

**Chief Technology Officer (CTO)**, *goodjourney.io* Feb 2023 - Present
Lead technology for goodjourney.io, building a ground transportation operating system and global supplier network that handles bookings, drivers, vehicles, and dispatch so rides can be booked anywhere in the world.

**Chief Product Officer**, *goodjourney.io* Dec 2022 - Jan 2023
Led product for goodjourney.io as the company began building a ground transportation operating system and global supplier network.

**Writer & Author**, *Kodeco (formerly raywenderlich.com)* Feb 2014 - Present
Turn complex mobile development concepts into practical tutorials and books that help developers build apps with iOS, Swift, and Flutter.

**Principal Software Engineer**, *Capital One* Mar 2017 - Aug 2019
Worked across iOS applications, backend microservices, identity and secure authentication services, Presto, and contactless payments and verification, while mentoring engineers and giving tech talks at Dev Days.

**Software Engineer**, *Capital One* Mar 2016 - Mar 2017
Built iOS interfaces for Capital One Wallet and the flagship app, and created the entire UI for T&EZ from scratch to help corporate travelers manage mileage, expenses, and receipts.

**Software Engineer**, *IBM · Austin, TX* Jul 2014 - Mar 2016
Developed enterprise mobile features for IBM Connections and IBM Verse across iOS and Android—including email management, contacts, calendars, and instant messaging—collaborating with product, design, and backend teams to improve responsiveness, reliability, and usability across multiple releases.

**iOS Developer Intern**, *Mindsense* Jan 2014 - May 2014
Built front-end features for Mail Pilot 2 on iPhone and iPad, contributing to the app's mobile email experience.

**Undergraduate Teaching Assistant · CS 3714, Mobile Development**, *Virginia Tech* Aug 2013 - Dec 2013
Helped students turn mobile development concepts into working iOS apps through project guidance, feedback, and hands-on programming assignments.

**iOS Developer Intern**, *Gannett* May 2013 - Aug 2013
Explored new iOS experiences for USA Today and built tools to capture and email console logs, helping developers investigate app issues.

## Honors & awards

**Capital One Identity Hackathon · First place**
Won first place for a liveness verification prototype that recorded randomly prompted facial gestures and detected multiple people in view to distinguish a live person from a static photo.

**VT Hacks · Fitnet award** Apr 2014
Won first place for the most creative app incorporating multiple workout motivators. Built Project Ham with a team: an iOS workout companion with scheduling, alerts, and a virtual hamster whose fitness reflects the user's workout habits.

## Patents

Co-inventor on three granted U.S. patents. Patents and related applications

### Systems and methods for transaction card-based authentication

Co-invented an NFC challenge-response method that authenticates a transaction card using a user's device.
Inventors: Kevin Osborn, Samuel Patrick Edwards, Vincent Vy Ngo
US 12,143,515 · Granted Nov 12, 2024

### Computing systems utilizing generated unique authorization identifiers for authorizing user operations and methods of use thereof

Co-invented a system that combines facial identity verification, unique authorization identifiers, location, and timestamps to authorize user operations.
Inventors: Lukiih Cuan, Dakyung Song, Vincent Vy Ngo
US 11,463,436 · Granted Oct 4, 2022
US 10,791,114 · Granted Sep 29, 2020

## Publications

2025

1. Data Structures & Algorithms in Swift
Kodeco · Book · 2018 - 2025
Co-authored a textbook that teaches the theory, implementation, and real-world application of data structures and algorithms in Swift, suitable for university courses and independent study. The Swift edition later served as source material for Kotlin and Dart adaptations by other authors.

2024

2. Flutter Apprentice
Kodeco · Book · 2020 - 2024
A hands-on guide that takes developers from Flutter fundamentals to building apps for iOS and Android from a single codebase.

2020

3. iOS Accessibility: Getting Started
Kodeco · Tutorial · 2016 · Updated 2020
An introduction to building more accessible iOS apps with VoiceOver and the Accessibility Inspector, originally published in 2016 and substantially updated in 2020.

2018

4. Swift Algorithm Club: Graphs & Graph Algorithms
raywenderlich.com · Tutorial series · 2017 - 2018
A four-part body of work covering adjacency-list graphs, depth-first search, Prim's minimum-spanning-tree algorithm, and Strassen's matrix-multiplication algorithm in Swift.

2017

5. Introduction to Protocol Buffers on iOS
raywenderlich.com · Tutorial · Feb 6, 2017
A practical guide to serializing structured data with Protocol Buffers and integrating the generated models into an iOS app.

2015

6. iOS 9 by Tutorials
Ray Wenderlich · Book · Oct 7, 2015
Co-authored a practical guide that helps developers turn the new iOS 9 APIs into features for their own apps.

7. How to Create an iOS Book Open Animation
raywenderlich.com · Two-part tutorial · Aug 11 - 12, 2015
A two-part, step-by-step tutorial for building a book-opening interaction with page flips and a custom collection-view layout.

2014

8. iOS 8 by Tutorials
Ray Wenderlich · Book · Sep 24, 2014
Co-authored a guide to adopting iOS 8 APIs through working examples in Swift and Objective-C.

9. NSScanner Tutorial: Parsing Data in Mac OS X
raywenderlich.com · Tutorial · Feb 27, 2014
A hands-on tutorial that shows developers how to turn text into structured data with NSScanner on macOS.

## Tech Talks

**Workshop One: Swift Collection Protocols**, *RWDevCon · with Kelvin Lau* 2018
Co-led a workshop on Swift collection protocols and the standard library, teaching advanced techniques for everyday development and technical interviews.

**Debugging, Optimizing & Fixing iOS Memory Leaks**, *RWDevCon* 2016
Presented techniques for using Xcode Instruments to diagnose memory leaks, debug iOS apps, and improve performance.

## Research

**Research Assistant · Part-time**, *Virginia Tech* Aug 2013 - Jun 2014
Contributed to Virtual Town Square, a project exploring online community interaction and civic engagement, with Dr. Andrea L. Kavanaugh and Dr. Manuel A. Pérez-Quiñones at Virginia Tech.

**Undergraduate Research Assistant**, *Mobile Software Engineering Lab · Virginia Tech* Nov 2012 - May 2013
Explored iOS 6 features with Dr. Osman Balci, built example apps and tutorials, and supported students through teaching assistant workshops, coursework creation, grading, and office hours.

## Skills & languages

Swift · Objective-C · Java · C · Python · React.js · iOS development · Mobile application development · Web development · Object-oriented design · Xcode · GitHub · Product development · Technical leadership · Software deployment
**English** · Native or bilingual proficiency
**Mandarin** · Limited working proficiency

## Interests

Hackathons · iOS development · Mobile · Golf · Music · Teaching · Writing